From the equation (9.14), pressure is equal to 2/3 of mean kinetic energy per unit volume.

9.2.4 Some elementary deductions from kinetic theory of gases

Boyle's law:

From equation (9.12), we know that $PV = \frac{2}{3}U$

But the internal energy of an ideal gas is equal to N times the average kinetic energy ($\in$) of each molecule.

$$U = N\in$$

For a fixed temperature, the average translational kinetic energy $\in$ will remain constant. It implies that

$$PV = \frac{2}{3} N\in \qquad \text{Thus} \quad \text{PV = constant}$$

Therefore, *pressure of a given gas is inversely proportional to its volume provided the temperature remains constant. This is Boyle's law.*

Charles' law:

From the equation (9.12), we get $PV = \frac{2}{3}U$

For a fixed pressure, the volume of the gas is proportional to internal energy of the gas or average kinetic energy of the gas and the average kinetic energy is directly proportional to absolute temperature. It implies that

$$V \propto T \text{ or } \frac{V}{T} = \text{constant}$$

This is Charles' law.

Avogadro's law:

This law states that at constant temperature and pressure, equal volumes of all gases contain the same number of molecules. For two different gases at the same temperature and pressure, according to kinetic theory of gases,

From equation (9.6)

$$P = \frac{1}{3}\frac{N_1}{V} m_1 \overline{v_1^2} = \frac{1}{3}\frac{N_2}{V} m_2 \overline{v_2^2} \qquad (9.15)$$

where $\overline{v_1^2}$ and $\overline{v_2^2}$ are the mean square speed for two gases and N_1 and N_2 are the number of gas molecules in two different gases.

At the same temperature, average kinetic energy per molecule is the same for two gases.

$$\frac{1}{2} m_1 \overline{v_1^2} = \frac{1}{2} m_2 \overline{v_2^2} \qquad (9.16)$$

Dividing the equation (9.15) by (9.16) we get $N_1 = N_2$

This is Avogadro's law. It is sometimes referred to as Avogadro's hypothesis or Avogadro's Principle.

9.2.5 Root mean square speed (v_{rms})

Root mean square speed (v_{rms}) is defined as the square root of the mean of the square of speeds of all molecules. It is denoted by $v_{\text{rms}} = \sqrt{\overline{v^2}}$

Equation (9.8) can be re-written as,

$$\text{mean square speed } \overline{v^2} = \frac{3kT}{m} \qquad (9.17)$$

root mean square speed,

$$v_{\text{rms}} = \sqrt{\frac{3kT}{m}} = 1.73\sqrt{\frac{kT}{m}} \qquad (9.18)$$

From the equation (9.18) we infer the following

(i) rms speed is directly proportional to square root of the temperature and inversely proportional to square root of mass of the molecule. At a given temperature the molecules of lighter mass move faster on an average than the molecules with heavier masses.

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